



Rationalizing Complex Number Expressions

Algebra 2 Worksheet · Grade 10-11

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Learning Objectives

- Rationalize rational expressions with imaginary numbers in the denominator
- Use complex conjugates to rationalize complex denominators
- Express rationalized complex numbers in standard form $a + bi$

Rationalize each expression and write your final answer in standard form $a + bi$.

1. Rationalize the expression and write in standard form.

$$\frac{4}{3i}$$

Answer: _____

2. Rationalize the expression and write in standard form.

$$\frac{2 + 3i}{5i}$$

Answer: _____

3. Rationalize the expression and write in standard form.

$$\frac{3}{2 - 5i}$$

Answer: _____

4. Rationalize the expression and write in standard form.

$$\frac{7}{2i}$$

Answer: _____

5. Rationalize the expression and write in standard form.

$$\frac{1 + i}{4i}$$

Answer: _____

6. Rationalize the expression and write in standard form.

$$\frac{5}{3 + i}$$

Answer: _____



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7. Rationalize the expression and write in standard form.

$$\frac{2}{1-i}$$

Answer: _____

8. Rationalize the expression and write in standard form.

$$\frac{4+i}{2-3i}$$

Answer: _____

9. Rationalize the expression and write in standard form.

$$\frac{6}{-2i}$$

Answer: _____

10. Rationalize the expression and write in standard form.

$$\frac{3-2i}{1+2i}$$

Answer: _____



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Remind students that $i^2 = -1$ and that multiplying a complex number by its conjugate eliminates the imaginary part.

Solutions

1. Rationalize the expression and write in standard form.

$$\frac{4}{3i}$$

- Multiply numerator and denominator by i .
- The numerator becomes $4i$ and the denominator becomes $3i$ squared.
- Replace i squared with negative 1 in the denominator to get negative 3.
- Simplify to obtain negative four-thirds i .

Answer: $-\frac{4}{3}i$

2. Rationalize the expression and write in standard form.

$$\frac{2+3i}{5i}$$

- Multiply numerator and denominator by i to eliminate i in the denominator.
- The numerator becomes $2i$ plus $3i$ squared and the denominator becomes $5i$ squared.
- Replace each i squared with negative 1.
- Simplify to get $2i$ minus 3 over negative 5.
- Rewrite in standard form as three-fifths minus two-fifths i .

Answer: $\frac{3}{5} - \frac{2}{5}i$

3. Rationalize the expression and write in standard form.

$$\frac{3}{2-5i}$$

- Multiply numerator and denominator by the conjugate 2 plus $5i$.
- Distribute in the numerator to get 6 plus $15i$.
- Multiply the conjugates in the denominator using the shortcut to get 4 minus 25 times negative 1 .
- Simplify the denominator to 29 .
- Write the result as six twenty-ninths plus fifteen twenty-ninths i .

Answer: $\frac{6}{29} + \frac{15}{29}i$

4. Rationalize the expression and write in standard form.

$$\frac{7}{2i}$$

- Multiply numerator and denominator by i .
- The numerator becomes $7i$ and the denominator becomes $2i$ squared.
- Substitute i squared equals negative 1 in the denominator to get negative 2 .
- Simplify to negative seven-halves i .

Answer: $-\frac{7}{2}i$



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5. Rationalize the expression and write in standard form.

$$\frac{1+i}{4i}$$

- Multiply numerator and denominator by i .
- Distribute i in the numerator to get i plus i squared.
- Replace each i squared with negative 1 to get i minus 1 over negative 4.
- Rewrite in standard form as one-fourth minus one-fourth i .

Answer: $\frac{1}{4} - \frac{1}{4}i$

6. Rationalize the expression and write in standard form.

$$\frac{5}{3+i}$$

- Multiply numerator and denominator by the conjugate 3 minus i .
- Distribute in the numerator to get 15 minus $5i$.
- Multiply the conjugates in the denominator to get 9 minus i squared.
- Replace i squared with negative 1 to get a denominator of 10.
- Simplify the fraction to three-halves minus one-half i .

Answer: $\frac{3}{2} - \frac{1}{2}i$

7. Rationalize the expression and write in standard form.

$$\frac{2}{1-i}$$

- Multiply numerator and denominator by the conjugate 1 plus i .
- Distribute in the numerator to get 2 plus $2i$.
- Multiply the conjugates in the denominator to get 1 minus i squared.
- Replace i squared with negative 1 to get a denominator of 2.
- Divide each term to obtain 1 plus i .

Answer: $1 + i$

8. Rationalize the expression and write in standard form.

$$\frac{4+i}{2-3i}$$

- Multiply numerator and denominator by the conjugate 2 plus $3i$.
- Use FOIL on the numerator to get 8 plus $12i$ plus $2i$ plus $3i$ squared.
- Replace i squared with negative 1 and combine like terms to get 5 plus $14i$.
- Multiply the conjugates in the denominator to get 4 minus 9 times negative 1, which equals 13.
- Write the result as five thirteenths plus fourteen thirteenths i .

Answer: $\frac{5}{13} + \frac{14}{13}i$



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9. Rationalize the expression and write in standard form.

$$\frac{6}{-2i}$$

- Multiply numerator and denominator by i .
- The numerator becomes $6i$ and the denominator becomes negative $2i$ squared.
- Replace i squared with negative 1 so the denominator becomes positive 2.
- Simplify to obtain $3i$.

Answer: $3i$

10. Rationalize the expression and write in standard form.

$$\frac{3-2i}{1+2i}$$

- Multiply numerator and denominator by the conjugate 1 minus $2i$.
- Use FOIL on the numerator to get 3 minus $6i$ minus $2i$ plus $4i$ squared.
- Replace i squared with negative 1 and combine like terms to get negative 1 minus $8i$.
- Multiply the conjugates in the denominator to get 1 minus 4 times negative 1, which equals 5.
- Write the result as negative one-fifth minus eight-fifths i .

Answer: $-\frac{1}{5} - \frac{8}{5}i$



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